

TASK 3: Task Automation (with Execution Output)

CodeAlpha Python Programming Internship • Project Submission

1. Source Code Implementation

```
import re

def extract_emails_automation():
    print("--- Email Extraction Automation ---")

    sample_data = """
    Hello team, please contact hr@codealpha.tech for internship queries.
    For technical support, reach out to support.tech@domain.com.
    John's personal email is john.doe123@gmail.org.
    """

    source_file = "source_text.txt"
    output_file = "extracted_emails.txt"

    with open(source_file, "w") as f:
        f.write(sample_data)

    try:
        with open(source_file, "r") as file:
            content = file.read()

        email_pattern = r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
        emails_found = re.findall(email_pattern, content)

        if emails_found:
            unique_emails = list(set(emails_found))
            with open(output_file, "w") as out_file:
                for email in unique_emails:
                    out_file.write(email + "\n")

            print(f"\nSuccess! Found {len(unique_emails)} unique email(s).")
            print(f"Saved directly to '{output_file}':")
            for em in unique_emails:
                print(f"- {em}")
        else:
            print("No email addresses found.")

    except FileNotFoundError:
        print(f"Error: The file '{source_file}' was not found.")

if __name__ == "__main__":
    extract_emails_automation()
```

2. Expected Terminal Output

```
--- Email Extraction Automation ---  
Created a dummy file 'source_text.txt' for testing.  
  
Success! Found 3 unique email(s).  
Saved directly to 'extracted_emails.txt':  
- hr@codealpha.tech  
- support.tech@domain.com  
- john.doe123@gmail.org
```